

ICD-FPTU: Benchmarking Image Capture Duplication Rate and F1-Score for Pothole Surveillance Systems

Hai Anh Nghiem Xuan*, Duy Dinh Tran*, Vinh Thanh Chu, Dang Hai Nguyen

Abstract—Real-time pothole surveillance system is crucial for supervising anomalies and automated road maintenance infrastructure. While being presented as a promising application, there are still many problems that prevent it from real-world production deployment. The previous literature has extensively studied dataset diversity, model optimization, and edge-hardware constraints, but the system architecture itself remains an overlooked bottleneck. In this paper, we introduce the Image Capture Duplication (ICD) problem, a critical issue in current architectures where a system continuously captures and uploads redundant images of the same pothole as a vehicle approaches it. To address this, a downstream filtering module is required to minimize data volume before transmission to the backend server. We introduce the ICD-FPTU dataset and related metrics (Precision system and F1) to benchmark standard YOLO-based detection against our proposed tracking-and-cooldown filtering pipeline. Our evaluation demonstrates that while standard YOLO setups achieve high Recall at the cost of massive data duplication and low Precision system, our integrated approach (YOLO combined with ByteTrack and optimized cooldown frames) reduces transmitted detections by 97.7% while improving system F1-score from 0.0208 to 0.3944 with only a moderate reduction in Recall under dense pothole road scenarios.

I. INTRODUCTION

Road transportation infrastructure serves as the fundamental backbone of modern socioeconomic mobility and logistics networks. Maintaining the structural integrity of this network is critical, as surface degradation directly threatens traffic safety, accelerates mechanical vehicle wear, and reduces ride comfort. Among various pavement surface failures, potholes represent one of the most hazardous localized structural anomalies, contributing significantly to traffic accidents and sudden evasive maneuvering risks [1]. Traditional continuous structural surveillance methodologies heavily depend on manual visual inspections or highly specialized laser-scanning inspection vehicles [2]. However, these conventional inspection techniques are inherently labor-intensive, cost-prohibitive, and logistically time-consuming, resulting in delayed maintenance interventions and prolonged exposure to hazardous road conditions.

To address these limitations, vision-based automated Pothole Surveillance Systems (PSS) have been widely proposed

in recent literature. The typical system architecture is generally structured as follows: an object detection model is deployed on an edge-camera mounted to a mobile vehicular platform (e.g., cars or motorbikes); when the model detects a pavement anomaly with a confidence score exceeding a predefined threshold, the corresponding frame is captured and transmitted back to a centralized backend server alongside its spatio-temporal metadata (such as GPS latitude and longitude coordinates) to populate a localized damage map (as illustrated in Fig 1). While this conceptual framework is highly promising for automated road monitoring and damage evaluation, bridging the gap between theoretical prototyping and real-world production deployment remains a challenge.

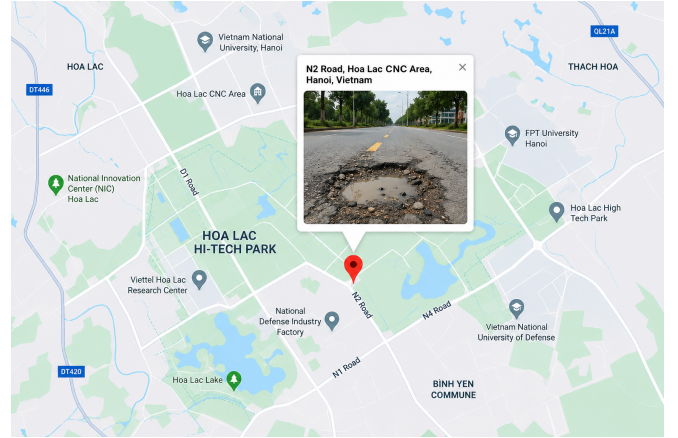


Fig. 1. Example of localized damage map.

Historically, transitioning this surveillance architecture to real-world production has faced three primary challenges: data scarcity across diverse road types and different weather or illumination conditions [3], the architectural challenge of detecting multiscale and highly variable small objects at speed [4], and the hardware limitations of resource-constrained edge-computing devices [5]. While former studies have optimized these components individually through robust deep learning models and lightweight edge network architectures, current literature consistently overlooks a fundamental bottleneck in the overall system execution logic: data transmission redundancy.

As shown in Fig. 2, the conventional data pipeline described by former researchers typically follows a direct loop: Camera → Frame Capture → YOLO Detection → Server [6], [7]. While theoretically promising, this pipeline fails in practical, mobile deployment environments. Depending on the camera's mounting position and view angles, a real-time

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detector will successfully identify the exact same physical pothole across dozens of consecutive frames from a distance before the vehicle actually passes it. This phenomenon leads to continuous, redundant image capturing and massive data duplication streams being transmitted to the backend server. For high-velocity or high-frame-rate systems, this Image Capture Duplication (ICD) problem results in severe network bandwidth exhaustion, inflates cloud storage costs, and adds immense processing overhead to the downstream system—especially when navigating roads with highly dense, concentrated potholes.

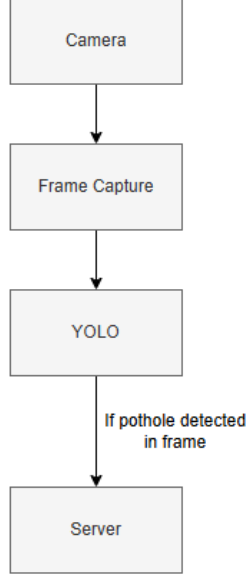


Fig. 2. Conventional data pipeline.

To resolve this system-level inefficiency, this paper introduces a deployment-ready architecture that integrates a downstream spatial-temporal filtering module designed to eliminate data duplication prior to cloud transmission (Fig. 3). The optimized pipeline processes data dynamically: Camera $\rightarrow$ Frame $\rightarrow$ YOLO + Filtering Module $\rightarrow$ Cloud Transmission. By filtering out redundant spatial events, this module drastically reduces the volume of uploaded images while preserving critical detection metrics.

To rigorously evaluate this paradigm, we introduce the ICD-FPTU dataset, a specialized benchmark designed to measure image duplication rates under varying environmental complexities, and system-level metrics (R_{ICD} , $Precision_{sys}$, $F1$) for evaluating surveillance efficiency. Utilizing this dataset and proposed metrics, we benchmark standard end-to-end YOLO detection architectures against our proposed filtering framework.

II. EXPERIMENT SETUP

To evaluate the performance of our proposed filtering framework against conventional detection baselines, this section describes the structural composition of the ICD-FPTU dataset and the formal mathematical criteria established to measure system-level redundancy.

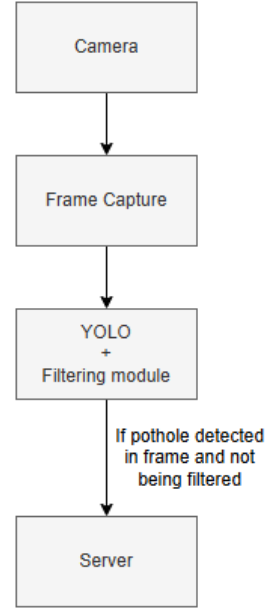


Fig. 3. Proposed data pipeline for Image Capture Duplication Reduction.

A. ICD-FPTU Dataset

Because the critical challenge of transmission redundancy in real-time automated anomaly surveillance has remained fundamentally unaddressed in previous literature, a key contribution of this study is the curation and formalization of the ICD-FPTU dataset [8]. This dataset is engineered specifically to provide a specialized evaluation benchmark for evaluating downstream spatio-temporal filtering frameworks built upon a YOLO-based baseline architecture.

1) *Operational Scenario Formalization:* Unlike standard computer vision problems with respect to illumination variance, weather extremes, or multiscale object features, the Image Capture Duplication (ICD) problem is fundamentally governed by the spatial density and frequency of pavement failures. To map these dynamics, our dataset formalization isolates two distinct operational road profile scenarios that uniquely influence both detection tracking robustness and system filtering efficiency:

- **Sparse Pothole Roads:** Routes characterized by isolated, geographically separated pavement anomalies. This subset establishes the baseline duplication metrics where a single tracking token is evaluated continuously across a linear line-of-sight vehicular approach vector (Fig. 4).
- **Dense Pothole Roads:** High-density, concentrated anomaly corridors where multiple structural defects manifest sequentially or concurrently within the camera's active field of view. This topology serves as the primary stress test for our filtering mechanism (Fig. 5).

As analyzed in later sections, these contrasting environmental profiles exert completely different mathematical strains on the aggregate system *Recall*.

2) *Dataset Curation and Metrics:* The ICD-FPTU dataset was compiled utilizing an aggregation of verified high-

Sparse Pothole Roads



Fig. 4. Example of Sparse Pothole Roads.

Dense Pothole Roads



Fig. 5. Example of Dense Pothole Roads.

resolution internet first-view driving and custom, self-recorded vehicular dashcam captures. In total, the dataset contains 7 comprehensive video streams comprising 6,125 frames and isolating 150 unique physical potholes. The exact quantitative distribution across the video registry is documented in Table I.

TABLE I
QUANTITATIVE PROPERTIES OF THE ICD-FPTU VIDEO REGISTRY

Video Identifier	Total Frame Count	Unique Physical Potholes
video_01	1,004	12
video_02	692	83
video_03	820	7
video_04	460	5
video_05	662	20
video_06	610	9
video_07	1,877	14
Total	6,125	150

3) *Spatio-Temporal Ground-Truth Annotation*: To properly validate tracking logic, standard static image annotation is insufficient; the dataset requires continuous temporal tracklets matching standard multi-object tracking specifications [9]. Consequently, we executed a meticulous, labor-intensive annotation process where every unique physical pothole is labeled across its entire lifespan—from its initial appearance frame to its exit frame—along with its corresponding pixel bounding coordinates (x_1, y_1, x_2, y_2) . A sample slice of this structural ground-truth serialization is illustrated in Table II.

This detailed telemetry mapping is critical due to the core operational logic of the surveillance pipeline. Because every raw sequential frame passes through the edge-YOLO extractor and the downstream filtering ruleset, a frame is only uploaded if an anomaly is present and not suppressed by the module logic. Retaining explicit frame-by-frame historical arrays enables flawless downstream matching, verification, and evaluation between distinct filtering parameters and true physical anomaly occurrences.

TABLE II

SAMPLE CONTINUOUS SPATIO-TEMPORAL GROUND-TRUTH LOGS IN
VIDEO_03 OF ICD-FPTU DATASET

Pothole ID	Frame ID	x_1	y_1	x_2	y_2
P1	71	193	113	237	128
P1	72	194	118	238	133
...	...	...	...	...	...
P1	290	354	737	473	838
P2	50	172	17	216	30
...	...	...	...	...	...
P2	330	93	692	306	839
...	...	...	...	...	...

B. Baseline Object Detector Implementation

We construct a baseline feature extractor, the underlying object detection module was trained on an aggregated corpus comprising 11,707 localized and driving-perspective images. This foundational training framework utilizes two complementary data domains to balance structural pattern learning with multi-viewpoint vehicular environmental challenges. The first training partition focuses on deep texture and structural pattern acquisition, utilizing a localized dataset containing 3,940 samples [10]. This subset consists primarily of centralized, close-up, high-visibility geometric captures of pavement failures under stable illumination profiles. This dataset provides an ideal baseline for the models to learn fundamental semantic features, including common pothole textures, structural degradation patterns, and interior surface variations. Conversely, the second training partition introduces intense environmental and perspective constraints. Sourced from the global Road Damage Detection (RDD) repository [11], this subset contributes 7,767 wide-angle images captured from authentic motorbike and vehicle dashcam frames. It presents critical deployment constraints, notably multi-scale distortions where anomalies appear highly condensed within the horizon line, and rapid structural movement perturbations.

The primary object detection framework was implemented using the recently proposed YOLOv26 [12] architecture to optimize real-time processing capabilities on hardware-bounded edge configurations. For this baseline benchmark evaluation, a lightweight YOLOv26n [13] was trained systematically across the hyperparameters detailed in Table III.

TABLE III
HYPERPARAMETER CONFIGURATION

Hyperparameters	Value
Input Resolution	$640 \times 640 \times 3$
Optimization Algorithm	AdamW
Batch Size	16
Total Training Epoch	100
Aggregated Training Volume	11,707 Images

C. Evaluation Metrics and System Formalization

While conventional computer vision frameworks evaluate models purely on frame-level bounding box overlap, optimizing an end-to-end PSS demands data-pipeline key metrics. Redundant transmissions directly translate to cloud

storage bloat and network bandwidth exhaustion. To properly quantify these system-level inefficiencies, we utilize a multi-tier evaluation framework.

Let N_d denote the total number of detected potholes that include duplicate instances across consecutive frames that are captured and transmitted to the cloud backend server during a surveillance run. Let N_{ud} denote the unique number of detected potholes that represent the actual count of distinct physical potholes successfully detected in N_d , and let N_{gt} represent the total number of physical potholes present along the inspected route.

1) *Surveillance Redundancy Framework*: To directly isolate data pipeline transmission overhead, we utilize our formulated Image Capture Duplication Rate (R_{ICD}) defined as:

$$R_{ICD} = \frac{N_d - N_{ud}}{N_d} \quad (1)$$

An idealized, completely optimized surveillance pipeline yields $R_{ICD} = 0$, implying that all captured images along a route isolate unique potholes, with each anomaly appearing in one and only one transmitted frame. Concurrently, we evaluate the System Uploaded Precision ($Precision_{sys}$) to represent the ratio of meaningful data captures relative to the total number of detected potholes:

$$Precision_{sys} = \frac{N_{ud}}{N_d} = 1 - R_{ICD} \quad (2)$$

Note that $Precision_{sys}$ differs from conventional object-detection precision and instead measures the proportion of unique uploaded detections among all uploaded detections.

2) *System Discovery and Operational Trade-offs*: Standard frame-by-frame YOLO-based detection pipelines maximize the system's ability to discover road failures, uploading thousands of images and resulting in a high System Recall ($Recall$). This metric represents the rate of unique ground-truth potholes the system successfully uncovers, calculated as:

$$Recall = \frac{N_{ud}}{N_{gt}} \quad (3)$$

Introducing an edge-filtering module inevitably introduces an operational trade-off: it aims to minimize data redundancy to achieve a lower R_{ICD} and a higher $Precision_{sys}$, but it risks the actual $Recall$ of our baseline YOLO system if an active filtering drops a pothole frame.

While R_{ICD} is an excellent metric to measure system redundancy, evaluating distinct filtering methods based purely on raw duplication counts can introduce bias due to the variance between N_{ud} and N_d . Therefore, to establish a unified benchmark parameter that fairly balances a framework's absolute anomaly discovery capacity against its transmission efficiency, we utilize the macro F1-score [14]:

$$F1 = 2 \times \frac{Recall \times Precision_{sys}}{Recall + Precision_{sys}} \quad (4)$$

By maximizing this consolidated $F1$ metric, the surveillance framework identifies the optimal configuration setting—keeping cloud-bound transmission lean without allow-

ing undetected critical structural anomalies to bypass the filter.

III. FILTERING METHODS AND BENCHMARKING

Prior to introducing the distinct algorithmic frameworks evaluated in this study, it is essential to formalize the evaluation data pipeline utilized to benchmark each method. The operational processing is visually systematically mapped in Fig. 6.

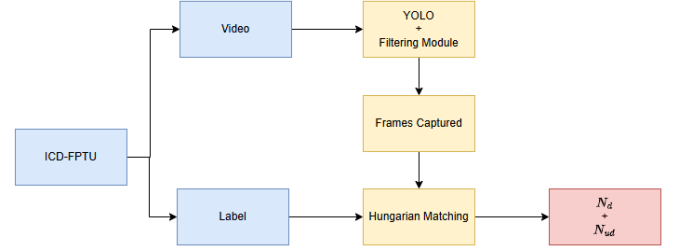


Fig. 6. Systematic block diagram of the data evaluation pipeline mapping structural video frames and telemetry annotations to generate system metrics via Hungarian optimization matching.

As illustrated, the sequential video frames from the ICD-FPTU dataset are fed continuously into the integrated pipeline consisting of the core YOLO object detector and the active downstream filtering module. The frame outputs that satisfy the specific threshold and suppression conditions of the active module are captured yielding the total uploaded detection count (N_d). To calculate the absolute unique anomalies successfully captured (N_{ud}), the predicted bounding box coordinates across all captured frames are matched back to the temporal ground-truth label tracking.

This matching optimization problem is solved using the Hungarian assignment algorithm [15], pairing predicted bounding boxes with unique ground-truth coordinates across consecutive frames. The assignment cost matrix is formulated based on the maximum Intersection over Union (IoU) metric calculated across a local temporal window denoted by t (where $t = 2$ in this study). Specifically, a detected bounding box in a given frame f is evaluated against all labeled ground-truth pothole coordinates spanning from frame $f - t$ to frame $f + t$. The algorithm extracts the maximum IoU value achieved against any true anomaly appearing within this temporal range if it exceeds 0.5; otherwise, it is set to zero; for any frame where other anomalies do not appear, the respective IoU coefficient is designated as zero. Once this temporal matching process is completed, the definitive values for N_d and N_{ud} are passed directly to the mathematical formulations detailed in Section II-C to generate the unified baseline metrics for the Image Capture Duplication Rate (R_{ICD}), System Uploaded Precision ($Precision_{sys}$), System Recall ($Recall$), and the aggregate system $F1$ -score.

A. YOLO-Baseline Performance Analysis

In YOLO pipeline configuration, every individual frame of a detected road distress event immediately triggers a frame capture to the backend server. As empirically expected,

evaluating this pure configuration against the continuous sequential streams of the ICD-FPTU dataset exposes the extreme structural data redundancy inherent to traditional Pothole Surveillance System (PSS) data pipelines.

The definitive quantitative results for this baseline evaluation are summarized in Table IV.

TABLE IV

SYSTEM PERFORMANCE BASELINE METRICS FOR UNFILTERED YOLO ARCHITECTURE

Records	N_d	N_{ud}	R_{ICD}	Recall	Precision	F1
4540	9,103	96	0.9895	0.6400	0.0105	0.0208

The empirical evaluation indicates that the standalone YOLO detector successfully identifies 96 unique anomalies out of the 150 true physical ground-truth pothole objects present along the inspected routes, yielding an absolute discovery limit of $Recall = 0.6400$. All subsequent downstream filtering modules introduced in this study will aim to constrain transmission volumes relative to this baseline to optimize efficiency, inherently risking drops in this maximum discovery value.

Crucially, this unfiltered setup captured 4,540 frame records (**Records** column in Table IV), generating an aggregate influx of $N_d = 9,103$ bounding box records uploaded to the cloud database. Because the network transmits a continuous wave of nearly identical frames as the vehicle approaches each physical asset, the resulting Image Capture Duplication Rate reaches an overwhelming $R_{ICD} = 0.9895$. Consequently, the System Upload Precision and overall System F1-Score collapse to a characteristically low $Precision_{sys} = 0.0105$ and $F1 = 0.0208$, respectively. This massive data footprint reinforces the argument that optimizing a production-ready pothole surveillance deployment requires look-ahead filtering architectural logic to minimize backend resource inflation without sacrificing baseline tracking recall bounds.

B. YOLO with Temporal Cooldown Filtering

A straightforward heuristic approach to mitigating high data transmission rates is the implementation of a temporal frame cooldown mechanism. Under this operational constraint, once a frame capture event is triggered by a positive pothole detection, a strict period defined by a set number of cooldown frames (f_{cd}) is enforced. Any subsequent positive detections occurring within this baseline window are systematically discarded by the system. While this mechanism prevents rapid data transmission, it introduces severe vulnerabilities in real-world driving environments. Specifically, if two distinct physical potholes are spatially clustered or if the vehicle is navigating at a high velocity, the fixed frame masking window will inadvertently suppress valid unique events, drastically degrading the system *Recall*.

To evaluate the operational limits of this approach, we benchmarked the pipeline across four incremental cooldown thresholds: $f_{cd} \in \{5, 10, 15, 20\}$. The definitive metrics generated from these iterations are consolidated in Table V.

TABLE V

SYSTEM PERFORMANCE METRICS UNDER VARYING FRAME COOLDOWN INTERVALS

f_{cd}	Records	N_d	N_{ud}	R_{ICD}	Recall	Precision	F1
5	986	1,898	89	0.9531	0.5933	0.0469	0.0869
10	519	989	83	0.9161	0.5533	0.0839	0.1457
15	355	663	74	0.8884	0.4933	0.1116	0.1820
20	274	508	71	0.8602	0.4733	0.1398	0.2158

The empirical results demonstrate a clear inverse relationship between the cooldown window length and data pipeline bloating. As f_{cd} escalates from 5 to 20 frames, the aggregate volume of captured frames drops sharply from 986 to 274 (compared to the unfiltered baseline of 4,540 frames), while the total transmitted detections (N_d) compress from 1,898 down to 508 (compared to the baseline ceiling of 9,103 detections). This systematic reduction bounds the duplication rate from $R_{ICD} = 0.9531$ down to $R_{ICD} = 0.8602$, yielding a corresponding increase in both System Uploaded Precision ($Precision_{sys}$) and the overall system F1-score. However, this transmission efficiency gain is fundamentally non-proportional to the steep deterioration observed in system discovery rates. The system *Recall* falls monotonically from the unfiltered YOLO baseline ceiling of 0.6400 down to 0.4733 at a 20-frame cooldown threshold. Crucially, even with this high degradation in anomaly discovery, the underlying frame records, transmission influx (N_d), and duplication overhead (R_{ICD}) remain unsustainably high.

C. YOLO with Integrated Multi-Object Tracking and Cooldown Filtering

To systematically resolve the structural trade-off between transmission inflation and anomaly discovery, we introduce a tracking-based filtering framework utilizing the ByteTrack architecture [16]. Rather than evaluating detections on isolated frames, this mechanism leverages a two-stage data association paradigm. It maps detection bounding boxes from high-confidence down to low-confidence thresholds across a temporal history, assigning unique tracklet tracking identifiers (IDs) to distinct physical anomalies. Under this control loop, a frame capture sequence is initialized if and only if a newly assigned tracklet ID manifests within the current frame view vector. To maximize downstream transmission compression, we hybridize this multi-object tracking logic with a fine-tuned micro-cooldown window ($f_{cd} \in [0, 20]$), mitigating tracklet fragmentation vulnerabilities. The quantitative results of this integrated framework are detailed in Table VI.

TABLE VI

SYSTEM PERFORMANCE UNDER INTEGRATED BYTETRACK AND COOLDOWN PIPELINE

f_{cd}	Records	N_d	N_{ud}	R_{ICD}	Recall	Precision	F1
0	257	270	72	0.7333	0.4800	0.2667	0.3429
5	240	240	70	0.7083	0.4667	0.2917	0.3590
10	205	205	70	0.6585	0.4667	0.3415	0.3944
15	174	174	61	0.6494	0.4067	0.3506	0.3765
20	153	153	56	0.6340	0.3733	0.3660	0.3696

The empirical telemetry demonstrates that migrating from static frame blanking to tracklet-based state tracking provides a substantial reduction in transmission overhead. Operating ByteTrack at a baseline zero cooldown ($f_{cd} = 0$) immediately slashes total uploaded detections (N_d) from the raw YOLO ceiling of 9,103 down to just 270, dropping the duplication rate to $R_{ICD} = 0.7333$. As shown in Table VI, adding micro-cooldown bounds of $f_{cd} = 10$ induces a small filtering penalty, dropping the unique physical anomaly count (N_{ud}) by merely two objects. However, it drives a major expansion in data thrift, compressing total transmitted detections down to 205 and reducing the replication factor to $R_{ICD} = 0.6585$. Consequently, the framework configured with a 10-frame micro-cooldown achieves an optimal operational balance, registering the highest overall system performance with an $F1$ score of 0.3944. Despite these stark engineering optimizations, this balanced architecture exhibits an inherent trade-off: its system *Recall* (0.4667) is roughly 17.33% lower than the unfiltered YOLO baseline ceiling, representing a missing index of 26 physical anomalies. This variance highlights explicit edge cases encountered by tracking-based data suppression models in mobile surveillance configurations. Before diving into a comprehensive evaluation across different routing scenarios, we emphasize that tracking-based data suppression architectures inherently encounter specific data-association anomalies that reduce the discovery of previously unseen potholes.

As explicitly illustrated in Fig. 7, identity switching (IDSW) and tracking fragmentation errors introduce notable artifacts into the data pipeline [17]. For example, as the vehicular perspective advances from frame 662 to frame 671, localized tracking fragmentation erroneously registering an active anomaly under a completely new identifier (where the distant pothole changes from ID₂₀₆ to ID₂₀₇). Although our integrated micro-cooldown mechanism robustly flags and suppresses duplicate updates, tracking identity re-assignment remains the primary failure mode limiting maximum anomaly discovery performance [18]. This systematic tracking vulnerability is documented across the high-density cluster sequence shown in Fig. 8. In frame 146, the local tracking index has already sequentially progressed up to ID₅₇. However, as the vehicle advances to frame 148, newly emerging, unlogged potholes appearing on the distant horizon are erroneously initialized with recycled historical tracklet identifiers (ID₄₂ and ID₅₆). Because the downstream data-suppression module recognizes these low-indexed tracking tokens as historical assets that have already triggered a server upload, the pipeline evaluates the newly discovered anomalies as pre-existing duplicates and discards the frames. This phenomenon generates omission errors within complex, spatially dense pothole clusters, directly causing a degradation in absolute system *Recall*.

IV. RESULTS ANALYSIS

To fully contextualize the performance boundaries of our optimized tracking-and-suppression pipeline relative to the standalone YOLO detector, this section provides an ablation



Fig. 7. Empirical visualization of tracking fragmentation and identity switching (IDSW) between frame 662 and frame 671, where an active historical pothole changes from ID₂₀₆ to ID₂₀₇.



Fig. 8. Visual documentation of ID re-assignment errors between frame 146 and frame 148. Emerging anomalies on the horizon erroneously inherit old tracking identifiers (ID₄₂ and ID₅₆), forcing the downstream filtering ruleset to drop the frames.

analysis across varying road complexities. As summarized globally in Table VII, while our hybridized framework significantly outperforms the baseline YOLO model across network-saving indices by saving 97.7% duplication detections in N_d and boosting $F1$ to 0.3944, it exhibits an absolute degradation of 26 missing unique physical anomalies (N_{ud}). To decouple this variance from tracking errors, we extensively evaluate performance across two distinct road profile domains: Sparse Pothole Roads and Dense Pothole Roads.

TABLE VII
GLOBAL PERFORMANCE COMPARISON: BASELINE VS. OPTIMIZED TRACKING

Method Configuration	Records	N_d	N_{ud}	$F1$
YOLO Baseline	4,540	9,103	96	0.0208
YOLO + ByteTrack + $10f_{cd}$	205	205	70	0.3944

A. Evaluation of Filtering Dynamics on Sparse Pothole Roads

To map framework robustness across isolated anomaly distributions, we isolate telemetry results from standard sparse pavement routes (video_01 and video_07). The comparative subset metrics are synthesized in Table VIII.

TABLE VIII
FILTERING EFFICIENCY UNDER SPARSE POTHOLE SCENARIOS

Video ID	Method Configuration	N_{ud}	N_{gt}
video_01	YOLO Baseline	9	12
video_01	YOLO + ByteTrack + $10f_{cd}$	9	12
video_07	YOLO Baseline	11	14
video_07	YOLO + ByteTrack + $10f_{cd}$	9	14

The empirical evaluation indicates that within sparse density topologies, our proposed tracking-filtering pipeline

matches the absolute discovery performance of the underlying YOLO baseline with a minor margin of loss (retaining identical discovery rates in video_01, and missing merely 2 objects in video_07). In sparse environments, maintaining near-maximum *Recall* is crucial, as municipal pavement management systems need detailed logging to track new, isolated road damage early, before it spreads and causes larger structural failures.

B. Evaluation of Filtering Dynamics on Dense Pothole Roads

Conversely, evaluating the system across high-frequency, tightly clustered anomaly profiles highlights a completely different set of operational dynamics. The performance telemetry across these dense scenarios (video_02) is compiled in Table IX.

TABLE IX
FILTERING EFFICIENCY UNDER DENSE POTHOLE SCENARIOS

Video ID	Method Configuration	N_{ud}	N_{gt}
video_02	YOLO Baseline	49	83
video_02	YOLO + ByteTrack + $10f_{cd}$	29	83

An analysis of the dataset tracking paths reveals that of the 26 total missing potholes logged in our dataset by tracking and cooldown compared to baseline, 20 specific omissions occur entirely within the complex and highly concentrated anomaly clusters previously documented in Fig. 8.

While from a purely numeric standpoint this *Recall* contraction appears severe, it represents a highly acceptable engineering trade-off for active production-grade deployments. Within a severely degraded road containing dozens of closely packed distress zones, the fundamental objective of an intelligent edge-surveillance vehicle shifts from counting every single structural anomaly to providing accurate spatial characterization of aggregate road decay. Although Recall decreases in dense pothole scenarios, the system still captures a substantial number of unique potholes (29 over 83), which may be sufficient for coarse-grained road-condition assessment and maintenance prioritization. Therefore, the drastic reduction in database record-bloat and network bandwidth conservation provided by our module outweighs the localized drop in absolute tracking recall.

V. CONCLUSION AND FUTURE WORK

In this study, we formalized a critical yet historically neglected bottleneck within automated computer vision pipelines: the Image Capture Duplication (ICD) problem in real-time Pothole Surveillance Systems (PSS). To anchor this challenge empirically, we introduced the ICD-FPTU dataset [8], providing a specialized multi-scenario benchmark with continuous spatio-temporal ground-truth annotation for downstream evaluation.

Through an extensive benchmarking campaign across multiple data-suppression strategies, our hybridized pipeline combining a YOLO object detector with the ByteTrack multi-object tracking architecture and a fine-tuned 10-frame

micro-cooldown ($f_{cd} = 10$) proved to be the optimal configuration. This framework established an exceptional balance between physical anomaly discovery and network bandwidth conservation, achieving the highest aggregate *F1* score of 0.3944. While the tracking system exhibits a localized drop in absolute *Recall* across dense pothole corridors due to identity re-assignment anomalies, we demonstrated that from an engineering system deployment perspective, the captured telemetry remains useful for infrastructure characterization of degradation.

Moving forward, we believe future research directions should focus directly on refining the multi-object tracking backend itself—specifically targeting data-association algorithms to eliminate ID re-assignment errors on emerging, undetected anomalies to structurally improve baseline recall limits. Furthermore, investigating alternative edge-filtering heuristics and multi-sensor fusion inputs will contribute significantly toward mature, field-ready deployments. Ultimately, we anticipate that the core architectural principles formalized in this study will transcend road infrastructure monitoring, paving the way for data capture efficiency, edge-optimized surveillance frameworks across diverse industrial inspection sectors.

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